UNIVERSITY OF ALBERTA

CMPUT 404 A1 (54143)

Fall 2025

WEB APPLICATIONS AND ARCHITECT

Territorial Acknowledgement

The University of Alberta respectfully acknowledges that we are located on Treaty 6 territory, a traditional gathering place for diverse Indigenous peoples including the Cree, Blackfoot, Métis, Nakota Sioux, Iroquois, Dene, Ojibway/ Saulteaux/Anishinaabe, Inuit, and many others whose histories, languages, and cultures continue to influence our vibrant community.

Instructor Information

Instructor
Dr. Hazel Campbell
Contact Information
Instructor Contact Form:
https://docs.google.com/forms/d/e/1FAIpQLSemXP7Bn1zcscmKhKYfK4kckTxyufNI5Fbmr-DJJcqFjyzJ9g/viewform?usp=dialog

Getting Help: https://uofa-cmput404.github.io/general/help.html
Office Hours: See next section.
Lecture Room & Time
Monday Wednesday Friday
1:00PM - 1:50PM

HC L1-L1
2-Sep-2025 - 8-Dec-2025
Course Page: https://uofa-cmput404.github.io/

Laboratory Coordinator(s)

Coordinator Name: Dr. Hazel Campbell
Email: hazelcam@ualberta.ca

Office Hours

Monday Wednesday Friday 2:10 PM to 2:50
Mondays; 2:10pm - 2:50pm - UCOMM **3-138**
Wednesdays; 2:10pm - 2:50pm - UCOMM **3-138**
Fridays; 2:10pm - 2:50 - **UCOMM 3-136**

If these times don't work for you, please email hazelcam@ualberta.ca for a Zoom appointment.

Teaching Assistant(s)

TA Name: TA contact information is on the Canvas

Zoom

Some lectures will also be available over Zoom.

- You must attend live.
- You will be required to attend in person for some lectures, all labs, all presentations, and all exams.

- You will be required to attend in person, on short notice, in the case of any technical difficulties with Zoom.
- You must be in Edmonton to take this course.
- You must attend Zoom as your real name (as displayed in Canvas).

Labs

You are required to have a laptop working for three hours.

LAB-D01-54144
Monday
5:00PM to 7:50PM
CSC B-10 (Check BearTracks for changes)

LAB-D02-54145
Wednesday
5:00PM to 7:50PM
CSC B-02 (Check BearTracks for changes)

All requests to change lab sections will be denied. **You must attend the lab section displayed in BearTracks.** You must change lab sections through BearTracks.

Labs and Project Parts are due at **16:00 (4PM) on Monday** before the Walkthrough. In the labs, the TAs will explain the submission process (typically online).

Calendar Description

Introduction to modern web architecture, from user-facing applications to machine-facing web-services. Topics include: the evolution of the Internet, relevant technologies and protocols, the architecture of modern web-based information systems, web data exchange and serialization, and service-oriented middleware. Prerequisites: CMPUT 291 and 301.

Additional Course Information (General Content)

This course introduces modern web architecture, ranging from user-facing applications to machine-facing web-services, through project-based learning. Topics include: multi-platform Web application frameworks utilizing SQL and NoSQL storage; HTTP/web architecture and protocols; integration with Web APIs; web data exchange and serialization.

Course Topics:

- The World-Wide Web: the tools and services.
- HTML
- CSS
- HTTP
- TCP/IP
- Web Frameworks
- Web Applications
- Web Services
- JavaScript
- XML
- REST
- Security mechanisms, authentication, encryption, firewall, Transport Security (TLS)
- AJAX
- WebSockets

Course Objectives/Outcomes

- Introduction to modern web architecture, from user-facing applications to machine-facing web-services.

- The course objective is to gain an overview of web-based technologies, with a software development focus on web services and web applications.

Required Course Materials

There is no required textbook for this course. Each module will cite online resources as required readings. You will find the links to each topic on the course page.
Link: https://uofa-cmput404.github.io/general/resources.html

Required Technology

Please contact the instructor before the add/drop deadline if you do not have access to the minimum technology recommended. The instructor will make arrangements for accommodating students who contact the instructor before this date. Failure to do so may result in a zero in any assessment that depends on the minimum technology.

Labs, Assignments, and Project

This course requires a Mac, Windows, or Linux laptop computer capable of:

1. Running an Ubuntu Linux VM or container.
    - Warning about RAM: Less than 16GiB RAM may not fit Windows 10/11 or macOS plus a VM.
    - Windows: Windows 11 running Ubuntu in WSL2. 16GiB RAM.
    - Intel Mac: VirtualBox running an Ubuntu VM. 16GiB RAM. CPU must support virtualisation acceleration.
    - ARM Mac: UTM running an Ubuntu VM. 16GiB RAM.
    - Linux: an Ubuntu Docker Container.
    - If you can not meet these requirements, you may need to use the Ubuntu Linux computers in the on-campus computer labs to test your work.
2. Running git and other software development tools, such as Visual Studio Code.

3. Working for the 3-hour duration of LAB sections. Your battery should be charged and last **longer** than 3 hours.

Although you may program in python at home or outside the lab, your implementations will be scored based on their behavior on a **standard Ubuntu Linux machine**, such as the CS department lab computers, or an Ubuntu VM. Firefox will be used for scoring frontend (HTML/CSS/JS).

Participation

A working smartphone, tablet, or laptop computer with an up-to-date web browser and working Wi-Fi.

Recommended Readings/Materials

Title: CMPUT 404 Recommended Reading
Link: https://uofa-cmput404.github.io/general/resources.html

Other Course Fees

Students will be required to purchase online hosting through Heroku in order to complete the coursework and course project. The expected fees are less than $50 USD, or approximately $70 CAD. If you are unable to pay these fees, you must contact your TA and instructor immediately. Failure to pay for Heroku in time to complete coursework will result in a mark of zero on the affected coursework.

Course Schedule

The weekly topic schedule is likely to be updated during the term; please see the live version.

On-Line Homework Disclaimer

Online homework is a component of this course and is provided by a third-party company. Please be aware that this company will be storing assessment information that may be associated with you. As a way to protect your personal information, you may be assigned a random ID to enter into the system. Only the random ID, performance on the online homework, and the affiliation to the University of Alberta for this random ID will be conveyed to the company. You are not required to provide any additional personal information to this company. If you have concerns about this, please contact the instructor of the course.

- GitHub Classroom (Microsoft) - You can create your own ID as long as it is not offensive. GitHub will be storing your labs, assignments and projects. GitHub will not have access to your marks, feedback, or grades.
- Heroku - Heroku will be storing your labs, assignments, projects. Heroku will not have access to your marks, feedback, or grades. See "Course Fees" above.
- Mentimeter - You will be assigned a random, non-identifying ID. The system does not require creating an account or payment. Do not enter your real name.
- Registration in the systems above and any monetary transactions are of your own accord and not the responsibility of the University.
- Students should be mindful of protecting their personal information and should be aware of how their personal information might be used and/or shared.
- Students MUST NOT use their @ualberta email address or CCID to register into the systems above, and instead should use a non-identifying email address or account.

Remote Delivery Considerations

Hybrid Synchronous Delivery

This course requires **live attendance** in-person. All labs, all exams, all presentations, and some lectures require in-person attendance.

- **Some Lectures** will be available in **hybrid synchronous** format. All lectures will be given in the lecture room on campus. At the same time, some lecture will also be streamed via Zoom and a recording of the lecture content will be made.
    - Zoom access is provided as a convenience only.
    - Zoom access is provided on a best-effort basis.
    - Students are expected to attend **during the lecture** either on Zoom or in-person.
    - Zoom access may be cancelled at any time.
    - You must be **available** to attend lectures in person.
        - You must be in Edmonton.
        - You must not have a scheduling conflict with this course.
        - Scheduling conflicts and travel **will not** be accommodated.
- In the case of **Emergency or Inclement Weather**, lecture will be provided via Zoom only.
- **Labs** must be attended **in-person only**.
- **Exams** must be taken **in-person only**. They will not be online. You must bring your Student ID (One Card).
- **Presentations and Guest Lectures** will be given **in-person only**.

To successfully participate in remote learning in this course, it is recommended that students have access to a computer with an internet connection that can support the tools and technologies the University uses to deliver content, engage with instructors, TAs, and fellow students, and facilitate assessments and examinations. If you encounter difficulty meeting the technology recommendations, please email the Office of Student Success and Experience (sse@ualberta.ca) directly to explore options and support. Please contact the instructor by the add/drop deadline [INSERT DATE APPROPRIATE FOR YOUR COURSE] if you do not have access to the minimum technology recommended.

Student Resources for Remote Learning:

Online learning may be new to you. Check out tips for success and find out more about online learning on the
Online + Remote Delivery Resources page.

Remote Proctoring Consideration:

Exams will be in person only.

Recordings of Synchronous Activities:

- Please note that class times for this course will be recorded. Recordings of this course will be used for the purposes of **review and for students with an excused absence only** and will be disclosed to:
  - other students enrolled in this semester of the class
  - TAs for this semester of the class
  - other CS Department Instructors
  - University Officials.
- You must attend this class **live, while the lectures are happening** unless you have an acceptable excuse (incapacitating illness, severe domestic affliction). You must be in Edmonton and available to attend in-person. If you are not available to attend in person do not take this course. Following University policy, we absolutely do not provide any accommodations for travel. https://calendar.ualberta.ca/content.php?catoid=56&navoid=18647#attendance "Unacceptable reasons include, but are not limited to personal events such as vacations, weddings, or travel arrangements. When a student is absent without acceptable excuse, a final grade will be computed using a raw score of zero for the work missed. Any student who applies for or obtains an excused absence by making false statements will be liable under the Student Academic Integrity Policy. Students should consult their Faculty for detailed information and requirements."
- Students have the right to not participate in the recording and are advised to turn off their cameras and audio prior to the recording, but students can still participate in the text-based chat. It is recommended that students remove all identifiable and personal belongings from the space in which they will be participating.

- Recordings will be made available until the start of the Final Exam and accessible in Canvas. Please direct any questions about this collection to the instructor of the course.

Home-based Lab Activities:

As part of the learning experience in this course, you will be required to undertake certain activities in or around your place of residence, To ensure that you undertake the activity safely and fully informed of the risks, please review the University of Alberta's Remote Learning Lab-Based Assignment Information Advisory. If you have questions or concerns, contact your instructor.

Grade Evaluation

Title	Assessment Type	Weight	Date of Evaluation
See next section.	Other		

Students must verify the date of the Final Exam on Bear Tracks when the Final Exam Schedule is posted.

Grades are unofficial until approved by the Department and/or Faculty offering the course.

Grade Evaluation

Grade Breakdown Table

What	Deliverable Type	Weight	Collaboration Policy	Due Date
Participation	Participation	8%	Solo Effort	Most Lectures
Language Assignment	Assignment	2%	Consultation	Tentative: **Thursday, September 18, 4PM**
Lab Assignments ×6	Assignment + Walkthrough	30% (5%×6)	Consultation	16:00 (4PM) on Monday Or next day if Holiday
Project Part 0	Teamwork	1%	Teamwork	Tentative: Thursday, September 21, 4PM
Project Part 1	Teamwork + Walkthrough	5%	Teamwork	Tentative: October 20 16:00 (4PM) on Monday
Project Part 2	Teamwork + Walkthrough	5%	Teamwork	Tentative: November 3 16:00 (4PM) on Monday
Project Part 3	Teamwork + Walkthrough	5%	Teamwork	Tentative: November 17 16:00 (4PM) on Monday
Project Part 4	Teamwork + Walkthrough	5%	Teamwork	Tentative: November 24 16:00 (4PM) on Monday
Project Part 5	Teamwork + Presentation	9%	Teamwork	Tentative: December 1 16:00 (4PM) on Monday

What	Deliverable Type	Weight	Collaboration Policy	Due Date
Peer Review x5	Google Form	-1/8 penalty if missing	Solo Effort	16:00 (4PM) day after Project Part 1, 2, 3, 4, Video
Final Exam	Exam (final)	30%	Solo Effort	See BearTracks
Helpful Pull Request	Extra Credit	+1%	Solo Effort	2nd to last week

- Please see the live schedule for exact due dates.
- See sections below under "Acceptable Collaboration" for details on each Collaboration Policy.

Participation & Language Assignment

See the individual work page.

Re-examination

There is no possibility of a re-examination in this course.

Final Letter Grades

University of Alberta Grading Policy

Grades reflect judgements of student achievement made by instructors and must correspond to the associated descriptor. These judgements are based on a combination of absolute achievement and relative performance in a class. Faculties may define acceptable grading practices in their disciplines. Such grading practices must align with the University of Alberta Assessment and Grading Policy and its procedures.

Letter Grades

Course Grades Obtained by Undergraduate Students:

Course Grades Obtained by Undergraduate Students

Descriptor	Letter Grade	Grade Point Value	Percentage	CS Department Interpretation
Excellent	A+	4.0	≥ 96.25%	Consistently original thinking that extends the material, demonstrated depth and breadth in the material, ability to integrate material with other subjects, ability to analyze and synthesize material at various levels of abstraction.
Excellent	A	4.0	≥ 91.25%	
Excellent	A-	3.7	≥ 87.5%	
Good	B+	3.3	≥ 83.75%	Like an A, but not consistent over time, or weak in a specific area.
Good	B	3.0	≥ 78.75%	
Good	B-	2.7	≥ 75%	
Satisfactory	C+	2.3	≥ 71.25%	Understand the core material but not its subtleties, can apply it to simple situations on own and to more complex situations with hints, evidence that the material has changed the way of thinking.
Satisfactory	C	2.0	≥ 66.25%	
Satisfactory	C-	1.7	≥ 62.5%	
Poor	D+	1.3	≥ 58.75%	Understands some of the core material but not its subtleties, can apply it to simple situations but often needs assistance, evidence that the material has had some change on the way of thinking.
Minimal Pass	D	1.0	≥ 50%	Shows some understanding of parts of the material, cannot apply it without some direction, little evidence that the material has changed the way of thinking.
Failure	F or F4	0.0	≥ 0%	Failure

Note: F4 denotes eligibility of a student to apply for a re-examination in a course.

Course Grades Obtained by Graduate Students:

This table reflects the GPA Point Value and Descriptor (e.g., Excellent, Good) for each Letter Grade.

Descriptor	Letter Grade	Grade Point Value
Excellent	A+	4.0
Excellent	A	4.0
Excellent	A-	3.7
Good	B+	3.3
Good	B	3.0
Satisfactory	B-	2.7
Satisfactory	C+	2.3
Failure	C	2.0
Failure	C-	1.7
Failure	D+	1.3
Failure	D	1.0
Failure	F	0.0

Access to Past or Representative Evaluative Material

Prior semester exams & review questions will be provided on Canvas. Some of it includes answer keys, most exam material does not include an answer key.

Statement of Expectations for AI Use

Summary

Is your answer English, drawing, diagram, etc?	Is your answer Code?
No GenAI allowed.	GenAI use must be cited
This must be completely your own work.	You must review, change, fix, and fully understand all code you submit.

In this course, we foster an equitable, inclusive, and accessible learning community that prioritizes individual critical, creative, and affective thinking as well as problem-solving skills specific to the discipline. To achieve the course learning outcomes, students must engage with tasks and assignments that allow them to develop and demonstrate their skills and knowledge authentically.

The use of AI tools is in this course is permitted for **computer program code only**. The use of AI tools is not allowed for tasks where the student is expected to respond in English, with a diagram, drawing, numeric result, mathematical formula, etc. Students are expected to complete non-code questions independently, without substantive assistance from others or GenAI.

You MUST not submit work by LLMs as your own, that is plagiarism. This also applies to other "AI" and Generative Models: ChatGPT, Lex, Page, DALL-E, Google Gemini, Microsoft Bing/Copilot, DeepSeek, Qwen, Llama, Cursor... and others. If you use LLMs you must cite it. This includes the corporation that made the AI, the AI, Subject, and Date. For example:

```
 // The following function from Microsoft, Copilot, "How do I write a merge sort
in JavaScript?", 2023-08-31
function mergeSort(array) {
```

If you use LLMs **you must cite it.**

If you use an integrated development environment with an LLM for completion (such as Copilot in VS Code), **you must cite it.**

```
 // The following written with completion assistance from Microsoft, Copilot,
2023-08-31
function mergeSort(array) {
```

You are expected to review, change, fix, and understand all code produced by GenAI. Your marks will be reduced (potentially to zero) if you cannot explain the code you submitted. You will be **graded** on your ability to understand your code.

Use of unmodified, unreviewed, unfixed GenAI output, or GenAI output that you do not understand may be considered **cheating** and a violation as outlined in the relevant

sections of the University of Alberta (September 2024) Student Academic Integrity Policy.

LLMs often produce the same answer. This will make it look like you copied (plagiarized) another student's answer! It is your responsibility to ensure the code you submit represents your individual work. If you submit the same code as another student, it will be considered plagiarism, even if the reason is you both used the same LLM.

LLMs like ChatGPT are wrong a lot. It does not understand computer science. It understands how to form sentences and paragraphs well enough to be convincing, but it doesn't actually understand what anything it is saying **means.** When it has the choice between two answers, with opposite meanings, it will pick the answer that looks more like things it has seen before, **not** the answer that is more **correct.** This means you need to double-check that what it tells you is actually correct.

User-driven websites such as Wikipedia and StackOverflow are always a better resource than GenAI, but of course if you use code from Wikipedia, StackOverflow or any other website, **you must cite it.** This is because other human programmers will usually check and downvote, remove, or fix bad information on StackOverflow. No one is checking the output of LLMs: if an LLM lies to you, no one will ever know.

ChatGPT and similar services are recording everything you tell it, and tracking you. Using ChatGPT/Bing/Bard etc. they are recording everything you say and how the LLM responds to you. There is no privacy.

- ChatGPT example from CMPUT 229:
  - Me: "What is the difference between the b and j pseudoinstruction in RISC-V?"
  - ChatGPT: "... The b instruction is actually a shorthand for the more general beq (branch if equal) instruction, which compares two registers and branches if they are equal. However, > > in the case of b, one of the registers is always x0 (the zero register), so the comparison > is always false, effectively causing an unconditional jump. ...the key difference between b > and j is that b is a relative branch instruction that jumps to a location within a limited range, while j is an absolute jump instruction that can jump to any address within the range of the program counter."

- There are 3 problems with this answer:
  1. b is not shorthand for beq: RARS replaces it with jal -- the actual replacement depends on the assembler used and the situation it is used in.
  2. This also makes ChatGPT's conclusion about relative limited range jumps and absolute jumps wrong. All jumps and branches in RISC-V are relative, short range jumps except jalr. RARS also (at least in every case I've seen) translates j to jal, not jalr, and jal is also a short-range, relative jump.
  3. If b gets translated to beq x0, Y, label then what guarantees that the other register isn't also zero? The comparison with beq x0, Y cannot always be false, despite what ChatGPT claimed. This really doesn't make any sense. It would make more sense for it to be translated to bne x0, x0, label ... but that's the opposite instruction of what ChatGPT claimed.

Relying on GenAI will lead to poor marks on the exam. This is because you simply will not have access to a computer during the exam.

The instructor may prohibit GenAI use for specific assessment tasks if it conflicts with course learning outcomes/objectives. Failure to abide by this guideline may be considered cheating and a violation as outlined in the relevant sections of the University of Alberta (September 2024) Student Academic Integrity Policy.

Please visit the University of Alberta Library's Using Generative AI for helpful information and suggestions about using AI ethically and creatively.

Missed Term Work/Final Exam Due to Non-medical Protected Grounds (e.g., religious beliefs)

When a term assessment or final exam presents a conflict based on non-medical protected grounds, students can register with the Academic Success Centre for accommodations via their Register for Accommodations website. Students can review their eligibility and choose the registration process specific for **Accommodations Based on Non-medical Protected Grounds**.

It is imperative that students review the dates of all course assessments upon receipt of the course syllabus, and register **AS SOON AS POSSIBLE** to ensure the timely application of the accommodation. Students who register later in the term may experience unavoidable delays in the processing of the registration, which can affect the accommodation.

Policies for Late and Missed Term Work

Late policy:

Absolutely no late work will be accepted. No extensions will be granted to individual students for any reason.

Missed Term Work: Assignments, Labs, Quizzes, Project Parts, Midterm Exams

A student who cannot complete Assignments, Labs, Quizzes, Project parts, or Midterm Exams due to incapacitating illness, severe domestic affliction, or other compelling reasons must contact the instructor within two working days of missing the assessment, or as soon as possible, to request an excused absence. If an excused absence is granted, then weight will be transferred to other course components (as determined by Canvas). An excused absence is a privilege and not a right. There is no guarantee that an absence will be excused. Misrepresentation of facts to gain an excused absence is a serious breach of the Student Academic Integrity Policy. In all cases, instructors may request adequate documentation to substantiate the reason for the absence, at their discretion.

> Unacceptable reasons include, but are not limited to personal events such as vacations, weddings, or travel arrangements. When a student is absent without acceptable excuse, a final grade will be computed using a raw score of zero for the work missed. Any student who applies for or obtains an excused absence by making false statements will be liable under the Student Academic Integrity Policy. Students should consult their Faculty for detailed information and requirements.

— https://calendar.ualberta.ca/content.php?catoid=56&navoid=18647#attendance

Missed Term Work: Participation

The 6 lowest marks for Participation will be dropped when calculating the course mark. No late Participation will be accepted. No Participation will be excused. Failure to complete Participation on time for any reason will result in a mark of zero. Please note that you can miss 6 instances of Participation without penalty.

Re-evaluation of Term Work

Re-Evaluation of Term Work follows the Computing Science department course policies. Any questions or concerns about marks on a particular assignment must be brought to the attention of the instructor *(not a TA)* within 10 calendar days of its return date. After that, we will not consider remarking or re-evaluating the work. So do not expect anyone to re-evaluate all the work you did all term long in the hopes of getting a higher final grade.

However, clerical errors such as incorrectly computing or recording a mark may be raised at any time prior to 2 working days following the final exam. It is the student's responsibility to confirm that their term work has been recorded properly.

Deferred Final Examination

A student who cannot write the final examination due to incapacitating illness, severe domestic affliction, or other compelling reasons can apply for a deferred final examination. Such an application must be made to the student's home Faculty Office within two working days of the missed exam and must be supported by appropriate documentation or a Statutory Declaration (see calendar on Attendance). Deferred examinations are a privilege and not a right; there is no guarantee that a deferred examination will be granted. The Faculty may deny deferral requests in cases where less than 50% of term work has been completed. Misrepresentation of facts to gain a deferred examination is a serious breach of the Student Academic Integrity Policy.

Academic Integrity and Student Conduct

The University of Alberta is committed to the highest standards of academic integrity and honesty, as well as maintaining a learning environment that fosters the safety, security, and inherent dignity of each member of the community, ensuring students conduct themselves accordingly. Students are expected to be familiar with the standards of academic honesty and appropriate student conduct, and to uphold the policies of the University in this respect.

Students are particularly urged to familiarize themselves with the provisions of the Student Academic Integrity Policy and the Student Conduct Policy, and avoid any behaviour that could potentially result in suspicions of academic misconduct (e.g., cheating, plagiarism, misrepresentation of facts, participation in an offence) and non-academic misconduct (e.g., discrimination, harassment, physical assault). Academic and non-academic misconduct are taken very seriously and can result in suspension or expulsion from the University.

All students are expected to consult the Academic Integrity website for clarification on the various academic offences. All forms of academic dishonesty are unacceptable at the University. Unfamiliarity of the rules, procrastination or personal pressures are not acceptable excuses for committing an offence. Listen to your instructor, be a good person, ask for help when you need it, and do your own work -- this will lead you toward a path to success. Any academic integrity concern in this course will be reported to the College.

Suspected cases of non-academic misconduct will be reported to the Office of Student Success and Experience. The College, the Faculty, and the Dean of Students are committed to student rights and responsibilities, and adhere to due process and administrative fairness, as outlined in the Student Academic Integrity Policy and the Student Conduct Policy. Please refer to the policy websites for details on inappropriate behaviours and possible sanctions.

The College of Natural and Applied Sciences (CNAS) has created an Academic Integrity for CNAS Students website. **To access this website, students must be signed in to their UAlberta account. Website content includes the importance of academic integrity, examples of academic misconduct and possible sanctions, and**

the academic misconduct and appeal process. Students can also access this material as an online, self-directed Canvas course and complete assessments to test their knowledge.

"Integrity is doing the right thing, even when no one is watching" -- C.S. Lewis

Contract Cheating and Misuse of University Academic Materials or Other Assets

Contract cheating describes the form of academic dishonesty where students get academic work completed on their behalf, which they submit for academic credit as if they had created it themselves. Contract cheating may or may not involve the payment of a fee to a third party, who then creates the work for the student.

Examples include:

1. Getting someone to write an essay or research paper for you.
2. Getting someone to complete your assignment or exam for you.
3. Posting an essay, assignment, or exam question to a tutorial or study website; the question is answered by a "content expert", then you copy it and submit it as your own answer.
4. Posting your solutions to a tutorial/study website, public server, or group chat and/or copying solutions that were posted to a tutorial/study website, public server, or group chat.
5. Sharing your login credentials to the course management system (e.g., Canvas) and allowing someone else to complete your assignment or exam remotely.
6. Using an artificial intelligence bot or text generator tool to complete your essay, research paper, assignment, or exam solutions for you (without the instructor's permission).
7. Using an online grammar checker to "fix" your essay, research paper, assignment, or exam solutions for you (without the instructor's permission).

Contract cheating companies thrive on making students believe that they cannot succeed without their help; they attempt to convince students that cheating is the

only way to succeed.

Uploading the instructor's teaching materials (e.g., course outlines, lecture slides, assignment, or exam questions, etc.) to tutorial, study or note-sharing websites, public servers, or chat apps is a copyright infringement and constitutes the misuse of University academic materials or other assets. Receiving assignment solutions or answers to exam questions from an unauthorized source puts you at risk of receiving inaccurate information.

Contract Cheating: CMPUT Courses

These are also contract cheating:

What	Where

What	Where

• Logging in as someone else
• Sharing your login credentials
• Sharing your anonymous ID
• Using someone else's anonymous ID
• Allowing someone else to log in as you
• Representing yourself as someone else
• Having someone else represent themselves as you
	• On other UAlberta services and linked services:
	  ◦ Zoom
	  ◦ gmail
	  ◦ Google Chat, Drive, ...
	  ◦ Lab computers
	  ◦ Wi-Fi
	  ◦ ...
	• On an external service, website, or app:
	  ◦ repository hosting services: GitHub, GitHub Classroom, Bitbucket, GitLab, ...
	  ◦ live quiz services: Mentimeter, ...
	  ◦ Textbook websites/apps
	  ◦ KnowledgeTree/MasteryGrids
	  ◦ online tutorials
	  ◦ online practice systems
	  ◦ online homework systems

- Misrepresenting authorship to a version control system such as git:
  - Forging git commit metadata (author, time, etc.)
  - Creating git commits where the author recorded did not create the changes being committed.
  - ...
- Submitting participation exercises for someone else.
- Representing yourself as someone else, or having someone else represent themselves as you to an instructor, TA, or other UA employee.
  - Attending lecture/lab/seminar for someone else.

    - Having someone else attend lecture/lab/seminar for you.

Intellectual Violence

In this course, *Intellectual Violence* is considered bullying. Intellectual violence is when one teammate uses their skill, knowledge, or experience, to intimidate or control the other teammate(s) rather than sharing and helping them learn. Examples of *Intellectual Violence*

- Using complex terminology or concepts to make others feel inferior.
- Dismissing or ridiculing colleagues' ideas or contributions.
- Withholding information to maintain a power imbalance.
- Creating an environment where only certain knowledge or skills are valued.
- Rejecting contributions without constructive feedback.
- Imposing overly strict code standards.
- Ignoring or delaying code reviews, pull requests, or commits.
- Favouring contributions from certain team members.
- Using harsh or condescending language in comments, code reviews, pull requests, issues, commit messages, ...
- Making contributors feel unwelcome.

Instances of *Intellectual Violence* may result in reduced marks and/or be referred to the Dean of Students under the Student Conduct Policy.

Appropriate Collaboration

Appropriate collaboration follows the CS course policy.

Students need to be able to recognize when they have crossed the line between appropriate collaboration and inappropriate collaboration. If students are unsure, they need to ask instructors to clarify what is allowed and what is not allowed.

Here are some tips to avoid copying on assessments:

1. Do not write down something that you cannot explain to your instructor.
2. When you are helping other students, avoid showing them your work directly. Instead, explain your solution verbally. Allowing your work to be copied is also considered inappropriate collaboration.
3. It is also possible that verbally discussing the solution in too much detail may result in written responses that are too similar. Try to keep discussions at a general or higher level.
4. If you find yourself reading another student's solution, do not write anything down. Once you understand how to solve the problem, remove the other person's work from your sight and then write up the solution to the question yourself. Looking back and forth between someone else's paper and your own paper is almost certainly copying and considered inappropriate collaboration.
5. If the instructor or TA writes down part of a solution in order to help explain it to you or the class, you cannot copy it and hand it in for credit. Treat it the same way you would treat another student's work with respect to copying, that is, remove the explanation from your sight and then write up the solution yourself.
6. There is often more than one way to solve a problem. Choose the method that makes the most sense to you rather than the method that other students happen to use. If none of the ideas in your solution are your own, there is a good chance it will be flagged as copying.

For programming assignments, powerful software tools are used to detect plagiarism. When the software tools indicate that there is similarity between two submissions, the submissions are reviewed by the instructor or teaching assistant. If the possibility that the standards for academic honesty were violated is confirmed, an investigation is started. Eventually the submitted solutions may be forwarded to the Faculty of Science Associate Dean of Students for further investigation and eventual sanctions.

- Each student must be able to verbally describe their exam answers and code, line by line, to a professor or TA, if asked to do so. Your mark may be reduced if you aren't able to explain your own work satisfactorily.

- For some substantial programming assignments and homework questions, students may discuss the concepts covered by the assignment with other students registered in the course as long as they do not share actual solutions or programming code.
- All suspected cases of plagiarism will be forwarded to the Dean's office and thoroughly investigated. Receiving a low mark for work not completed is a far superior alternative to this process and its possible long-term consequences for your career.

All suspected cases of plagiarism and other forms of cheating are immediately referred to the College of Natural and Applied Sciences (CNAS). CNAS, not your instructor, will determine what course of action is appropriate. We do not hesitate to send ALL cases of cheating to CNAS. Please do not put yourself or us into such an unpleasant situation. Please read the Student Conduct Policy carefully.

Appropriate Collaboration: Citation

If you include code or ideas from someone who isn't you (including from a Generative AI or LLM) you must cite it. Here is an example of an appropriate citation:

```
 # The following function from Django Software Foundation and contributors,
"Writing your first Django app, part 2", 2024-01-05
class Choice(models.Model):
 question = models.ForeignKey(Question, on_delete=models.CASCADE)
 choice_text = models.CharField(max_length=200)
 votes = models.IntegerField(default=0)
```

Example:

```
 # From https://stackoverflow.com/a/952952 by Boris Verkhovskiy, Downloaded
2023-05-04
flat_list = [item for sublist in l for item in sublist]
```

Appropriate Collaboration: Solo Effort: Participation, Quizzes, Exams

Participation exercises fall under the departments *Solo Effort* model unless announced by the instructor. *Solo Effort* must be completed by the student registered in the course without external assistance from any individual or organization.

- Students may only submit work authored by themselves.
  - Students **may not** submit work found online or generated by a large language model (LLM).
- Students **may not** discuss or exchange solutions, steps, strategies, code, links, code, images, videos, output, comments, repositories, answers, etc.
- Students **may not** consult with other students on how to solve the problem, unlike the consultation model described below.
- Students **may not** submit a participation exercise, quiz, or exam without attending the relevant lecture, lab, seminar, or exam.
- Students **may not** share a participation exercise, quiz, or exam link (URL).
- Students **may not** represent themselves as someone else, (or have someone else represent themselves as the student) during lectures, labs, seminars, or exams.
  - Including in-person or over Zoom, or any other remote video, voice call, chat, or email service.
  - See Contract Cheating above.

Violation of the above will be considered a breach of the Student Academic Integrity Policy.

Appropriate Collaboration: Confidential: Exams

Midterm and Final Exams are also *Confidential* in addition to *Solo Effort* as listed above.

- Students **may not** discuss the contents of the exam, except with instructors.
  - Students are not always able to take the exam at the same time, so do not discuss the contents of the exam even if you have already taken it!
- No human, computer, electronic assistance is allowed of any sort, including AI chatbots, calculators, tutors, etc.

Also see the Exam Conduct section below.

Appropriate Collaboration: Consultation: Assignments, Labs

Individual assignments and labs are under the department's *Consultation* model. That means you may discuss the labs with others, but you must create and submit a solution that is entirely your own work. If you consult with other students, you must list their names in a comment at the top of your submission or in your repository `README`, along with a brief description of the part(s) of the assignment you discussed.

How to consult with other students without plagiarizing:

- Study the Course, Computing Science Department, CNAS, and University policies listed above regarding Academic Integrity.
- All sources used must be cited.
  - If you use code snippets you must cite them, please see the examples above.
    - This includes code snippets that are exact copies (copy and paste),
    - as well as code snippets that are adapted from an online or LLM source.
  - All sources of information used, e.g., books, websites, students you talked to etc., must be cited in your submitted file or repo `README` for each assignment.
  - If student A cites student B, then B should also cite A as consulted.

- Individually develop your own solution for assignments and exercises. **Submit only your own work for evaluation.**
  - Each student is responsible for what is handed-in and must be able to explain it.
  - Students may only submit work authored by themselves. Work submitted by a student that is the work of someone else (e.g. another student or a tutor) either in part or in entirety is considered plagiarism.
- You can freely discuss the steps and solutions with your classmates on a conceptual verbal level.
- Limit discussion to be among students taking the current course, not students who took in earlier terms or other students. Consultation is a two-way process that benefits both sides.
- **Do not** exchange any text, code, images, videos, output, comments, repositories, or detailed (low-level) step-by-step procedures.
  - **Do not** share solutions.
- **Do not** give other students access to your solutions and do not seek access to other's solutions. This is considered plagiarism.
  - Do not show your code to classmates.
  - Do not look at a classmate's code.

Examples of consultation:

- Acceptable consultation example:
  - Student A has a problem with the code
  - Student A asks Student B for help
  - Student B explains the steps, concepts, or techniques they used to get their code working
  - Student A understands the fix
  - Student A can reproduce and explain the fix.
  - Student A submits the code
- Acceptable consultation example:
  - Student A needs to make a grid for a board game program.

    - Student A asks Student B for help.
    - Student B explains that they "used two nested for loops, one for the vertical and one for the horizontal."
    - Student A implements two nested for loops using their own unique code.
    - Student A can use the for loops to fix similar problems, and they can explain why each piece of code is needed, along with how it works to solve the problem.
    - Student A submits the code
- **Unacceptable** consultation example:
    - Student A needs to make a grid for a board game program.
    - Student A asks Student B for help.
    - Student B sends the code they used to make the board game grid for their solution.
    - Student A copies Student B's code into their own solution, changing it a little.
    - Student A submits the code
- **Unacceptable** plagiarism example:
    - Student A has a problem with the code
    - Student A asks Student B for help
    - Student B provides Student A with the code
    - Student A submits the code

Appropriate Collaboration: Teamwork: Group Projects

Within the team, code may be freely shared and worked on. Between two teams or two students in different teams, teamwork follows the collaboration model above: discussion but no sharing of solutions.

- As long as you are a part of a group, you are responsible for everything in the group project, whether you participated in every component or not.
- A group may only submit work authored by group members or appropriately cited and credited code that does not violate the author's license.

Exam Conduct

Please refer to the Examinations section of the Academic Calendar for more details on Conduct of Exams.

1. Do not start the test or open the test booklet until instructed.
2. If you are not on the class list, you may not write this examination.
3. If you do not present your Student ID, you may not write this examination.
    1. Your student photo I.D. is required at exams to verify your identity.
4. If you enter more than 30 minutes after the start of the examination may not write this examination. You must
   remain seated until 30 minutes has passed.
    1. Students must arrive at the specified time to take the exam. Once the exam has started, students must remain in the physical in-person or remote environment for at least 30 minutes. Students who arrive more than 30 minutes late for an in-person exam will not be permitted to take the exam. Students who arrive more than 30 minutes late for an online exam may have their exam attempt removed or disqualified by the instructor. In both cases students may apply for a deferred examination.
5. If you are sick, please do not attend the examination. Instead, apply for a deferred exam with your Faculty (e.g.
   Faculty of Science).
6. No electronic devices are allowed. This includes calculators, cell phones, smartwatches, and headphones. You may
   not have them near you, in your pocket, or take them to the washroom. If you brought electronics with you, you
   must leave them in your bag at the front of the classroom. If you are distracted

by the sound of other
students, bring ear plugs, not ear buds.

    1. You must leave your bag at the front of the classroom. You may be able to see it, but it must be out of reach.

7. You may write on the exam if you need scratch paper.
8. No notes are allowed, except 1 handwritten letter-sized pages prepared by you.
9. Place your student ID number on every page. If your exam pages come apart we will need this!
10. Multiple-choice questions: Please circle the correct answer.
11. For written-answer questions, please limit your answers to 1-2 sentences („30 words).
12. Print clearly, if I cannot read your handwriting you will lose marks. Do not write cursive.
13. For short answer questions, please give concrete examples. Tell me what things/people/groups/apps are named.
    Give real functions, real kinds of apps, describe a real test that might really exist. They can be made up/imaginary,
    but they must be specific examples that mention specific things.

    1. Bad: The application is a communication app. “As a user, I want to be able to communicate.” So the app
       needs to implement features that allow a good user experience for communicating.
    2. Good: The application is a secure voice chat called YellingWithFriends. “As a friend, I want my friends to be
       able to hear me.” YellingWithFriends needs to make sure all the friends can be heard, so a separate automatic
       volume is applied for each incoming audio stream.

14. When you are done, please hand in the entire exam, and your notes. You must hand in the entire exam, including
    pages which do not have questions on them. You are not allowed to keep a copy of any part of the exam or your
    notes.

15. Ensure you have signed the class list. A TA will be walking with the class list. The TA will check your student ID
    and allow you to sign the class list.
16. If you do not stop writing the exam when the instructor or proctor announces the end of the exam, your exam will not be graded, and you will receive a mark of zero.

Lecture Conduct

- Cell phone audio/ringers are to be turned off during lectures, labs and seminars.
- Cell phone and computer use during the lecture is only permitted for participation, note-taking, and other activities directly related to the lecture.
- Students who prevent the lecture from continuing will be asked to leave.

University Policy

Withdrawals

See the University Calendar for the relevant add/drop deadlines for each term.

Course Outlines

Policy about course outlines can be found in the Academic Regulations, Evaluation Procedures and Grading section of the University Calendar.

Student Academic Integrity

The University of Alberta is committed to the highest standards of academic integrity and honesty. Students are expected to be familiar with these standards regarding academic honesty and to uphold the policies of the University in this respect. Students are particularly urged to familiarize themselves with the provisions of the Student Academic Integrity Policy and the Student Conduct Policy (on the University of Alberta Policies and Procedures Online (UAPPOL) website) and avoid any behaviour which could potentially result in suspicions of cheating, plagiarism, misrepresentation

of facts and/or participation in an offence. Academic dishonesty is a serious offence and can result in suspension or expulsion from the University.

Recordings

Audio or video recording, digital or otherwise, of lectures, labs, seminars or any other teaching environment by students is allowed only with the prior written consent of the instructor or as a part of an approved accommodation plan. Student or instructor content, digital or otherwise, created and/or used within the context of the course is to be used solely for personal study, and is not to be used or distributed for any other purpose without prior written consent from the content authors.

Accommodations for Students

In accordance with the University of Alberta's Accommodation Policy and Discrimination and Harassment Policy, accommodation support is available to eligible students who encounter limitations or restrictions to their ability to perform the daily activities necessary to pursue studies at a post-secondary level due to medical conditions and/or non-medical protected grounds. Accommodations are coordinated through the Academic Success Centre, and students can learn more about eligibility on the Register for Accommodations website.

It is recommended that students apply **AS SOON AS POSSIBLE** in order to ensure sufficient time to complete accommodation registration and coordination. Students are advised to review and adhere to published deadlines for accommodation approval and for specific accommodation requests (e.g., exam registration submission deadlines). Students who request accommodations less than a month in advance of the academic term for which they require accommodations may experience unavoidable delays or consequences in their academic programs, and may need to consider alternative academic schedules.

Student Supports

Faculty of Science Student Services: The Faculty of Science Student Services office is located on the main floor of the Centennial Centre for Interdisciplinary Sciences (CCIS). This office can assist with the planning of Your Academics, and provide

information related to Student Life & Engagement, Internship and Careers, and Study Abroad opportunities. Please visit Advising for more information about what Faculty Academic Advisors can assist you with.

Academic Success Centre: The Academic Success Centre provides professional academic support to help students strengthen their academic skills and achieve their academic goals. Individual advising, appointments, and group workshops are available year round in the areas of Accessibility, Communication, Learning, and Writing Resources. Modest fees may apply for some services.

Feeling Stressed, Anxious, or Upset? It's normal for us to have different mental health experiences throughout the year. Know that there are people who want to help. You can reach out to your friends and access a variety of supports available on and off campus at the Need Help Now webpage or by calling the 24-hour Distress Line: 780-482-4357 (HELP).

Student Self-Care Guide This Self-Care Guide, originally designed by the Faculty of Native Studies, has broader application for use during students' learning. It provides some ideas and strategies to consider that can help navigate emotionally challenging or triggering material.

Learning and Working Environment

The Faculty of Science is committed to ensuring that all students, faculty and staff are able to work and study in an environment that is safe and free from discrimination, harassment, and violence of any kind. It does not tolerate behaviour that undermines that environment. This includes virtual environments and platforms.

If you are experiencing harassment, discrimination, fraud, theft or any other issue and would like to get confidential advice, please contact any of these campus services:

- Office of Safe Disclosure & Human Rights: A safe, neutral and confidential space to disclose concerns about how the University of Alberta policies, procedures or ethical standards are being applied. They provide strategic advice and referral on matters such as discrimination, harassment, duty to

accommodate and wrong-doings. Disclosures can be made in person or online using the Online Reporting Tool.

- University of Alberta Protective Services: Peace officers dedicated to ensuring the safety and security of U of A campuses and community. Staff or students can contact UAPS to make a report if they feel unsafe, threatened, or targeted on campus or by another member of the university community.
- Office of the Student Ombuds: A confidential and free service that strives to ensure that university processes related to students operate as fairly as possible. They offer information, advice, and support to students, faculty, and staff as they deal with academic, discipline, interpersonal, and financial issues related to student programs.
- Office of Student Success and Experience: They can assist students in navigating services to ensure they receive appropriate and timely resources. For students who are unsure of the support they may need, are concerned about how to access services on campus, or feel like they may need interim support while they wait to access a service, this office is there to help.

On-Campus Computer Labs

See the Undergraduate Conditions of Use and Lab Policies for information about undergraduate labs.

Disclaimer

Any typographical errors in this syllabus are subject to change and will be announced in class and/or posted on the course website. The date of final examinations is set by the Registrar and takes precedence over the final examination date reported in the syllabus.

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